



TEST & MEASUREMENT

Summary

- Technologies and processes are crossing industry boundaries, creating both pain and potential for test leaders
- Test strategies built on closed and proprietary methods put organisations at risk
- Partnerships with multi-industry companies provide the insight needed to adapt test organisations in time

uct innovation, the analogy can also be applied to test strategies.

A clear example of this relates to functional safety in the heavy manufacturing industry, where a process named IEC 61508 was created to prove the functional safety of embedded electronics. This was based on many years of learning and safety-criticality of products, and was later adopted by industries such as rail and automotive (EN 50126 and ISO 26262, respectively), where similar requirements were seen. Learning from experts in these standards can save time when adding functional safety testing to test strategy if, and when, it becomes necessary.

An added benefit that convergence provides is multi-industry resource pooling where vendors can now serve multiple industries simultaneously due to the alignment of their functional needs over time. In test, platform-based vendors can increase their industry-agnostic investment in things like processors or analogue-to-digital converters to provide better quality products at lower prices to all industries. Such investments made in hardware, software or services for test, multi-industry solutions, as opposed to single-industry options, maximise the opportunity for technology leverage.

Overcoming the cost of convergence

Despite its potential upside, convergence comes with its fair share of concerns as well, especially for test managers. Primarily, it adds complexity and demands the creation of more adaptable test platforms and even more flexible organisations.

Exchange of technologies between industries requires a concurrent update in testing and expertise in these newly-created hybrid domains. Automotive hybrid powertrains, for example, now require systems that can test controls, mechanics, thermodynamics, electronics,

software and even battery chemistries.

These requirements have led to the obsolescence of even the most-recently-made tests systems due to their inflexible, closed and proprietary platforms. Changes that must be made today revolve around the use of more open and modular hardware and software that work across I/O types, programming languages and vendors, along with well-defined APIs and interoperability standards.

This is exacerbated by the inability of organisations to visualise a way forward, as the future in a converged world is much more hazy. Companies, test strategies and test platforms must be designed keeping future trends and requirements in mind.

For example, aerospace companies, which have historically moved very conservatively and relied on long product life cycles, now need to be nimbler as their supply chain grows more closely tied to that of consumer electronics. As a result, testers at aerospace test organisations must contend with a much faster technology refresh rate and design test architectures that are adaptable to these changes. A potential solution for this would be to attend cross-industry networking events and to monitor other industries' trade publications.

Collaborating with organisations that have multi-industry experience is another avenue that can be followed as they learn to absorb unforeseen circumstances more effectively and leverage best practices from other industries. Important problems can, thus, be outsourced to more cognizant third parties, or strategic partnerships can be looked at in other industries around imminent trends like 5G and the Internet of Things (IoT).

NVIDIA and Audi partnering to accelerate technology development or Boeing and Embraer collaborating to take market share from competitors are just two of the many examples of how this type of cooperation can lift organisations above their industry peers. Re-evaluating where test happens in the supply chain and reviewing suppliers are also smart tactics. By being proactive, organisations can be prepared for what is next and maybe even influence it. **EFY**



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